

Question 1: (15Mks.)

- a. Identify the phases of compilation. 5mks
- b. Explain the role of each identified phase in (a) above. 5mks

Question 2: (15Mks.)

- a. Discuss the major challenge to compiler design? 5mks
- b. If String_1 and String_2 and String_3 are three strings defined as
String_1 = Compiler
String_2 = Construction
String_3 = Company
 - i. What is String_2.String_3?
 - ii. What is | String_2.String_3 | ?
 - iii. Give the singular proper prefix of String_2 that makes a valid word. 5mks

Question 3: (10Mks.)

Explain

- i. Finite Automata
- ii. Differentiate between Deterministic Finite Automata (DFA) and Non-deterministic Finite Automata (NFA).

Question 4: (10Mks.)

Given the finite automaton described as follows;

$M = (\{\text{Liquid, Solid, Gas}\}, \{\text{Heat, Cold}\}, \delta, \text{Liquid}, \{\text{Solid}\})$

Draw the automaton M and indicate the δ .